

Measuring What Was Said, Not Only How It Sounded

A hybrid deterministic and generative architecture for automated assessment of children's impromptu speech

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Abstract

We describe the architecture of a system that assesses short impromptu and prepared speeches by children aged 6 to 14 and returns coaching feedback. The contribution is architectural rather than algorithmic: every component uses established methods, and the claim concerns their composition. The system partitions assessment along a strict boundary. Delivery — speaking rate, hesitation, disfluency, repetition, lexical range, fundamental frequency variation and amplitude contour — is computed deterministically from the transcript's word-level timestamps and from the audio signal. Content is assessed by a large language model constrained to a seven-dimension rubric and given the transcript only; it is instructed explicitly that it cannot hear the audio. The corrective action returned to the child is selected by a fixed priority ordering evaluated in code, not chosen by the model. Acoustic measures are normalised within-subject against the speaker's own first three sessions and are withheld entirely until that baseline exists. No numeric score is shown to the child at any point.

This paper reports no empirical results. The system is implemented and operational, but word error rate on the target population has not been measured, agreement with human raters has not been computed, and no user study has been conducted. Section 9 states an evaluation plan in the future tense. Section 10 states the limitations this imposes.

1. Problem statement

1.1 Target population

The system targets children aged 6 to 14 preparing for, or practising the skills assessed in, school elocution, declamation, extempore and Just-A-Minute (JAM) formats. The implementation partitions this range into three bands — 6-8, 9-11 and 12-14 — and the topic corpus is filtered by band. The corpus contains 333 prompts distributed across those three bands, alongside a 1,046-token common-word list used for lexical rarity measurement.

Two roles are distinguished throughout the implementation. The *speaker* is the child. The *account holder* is a parent or guardian, who is the party that consents, and the party to whom numeric assessment is shown.

1.2 Three constraints

Acoustic heterogeneity. Fundamental frequency in this age range varies substantially with developmental stage. A single fixed threshold applied to pitch or amplitude will classify younger speakers differently from older ones for reasons unrelated to speaking skill. Any absolute acoustic criterion therefore encodes a confound between age and performance.

ASR degradation on child speech. Automatic speech recognition is trained predominantly on adult speech. Children exhibit higher fundamental frequency, different formant structure, greater articulatory variability and markedly more disfluency. Accented child English compounds this. Every downstream content judgement is conditioned on transcript quality, so transcription error is not an isolated component risk but a systemic one. We have not measured this error rate (Section 9.1).

Behavioural sensitivity to negative feedback. The system's objective is practice volume: a child who records regularly improves, and a child who stops recording does not. Assessment output that reduces the probability of a subsequent session is counterproductive regardless of its accuracy. This is a design premise, argued in Section 3.4 and Section 7.3 from reasoning rather than from measurement within this work.

2. Related systems

2.1 Adult speech coaching

A mature category of consumer tools assesses adult presentation and interview speech, typically reporting speaking rate, filler-word counts, and a composite described as a confidence or delivery score. These systems generally assume a speaker who has volunteered for evaluation, tolerates a numeric verdict, and falls within adult acoustic norms.

2.2 Child pronunciation and literacy

Systems addressing children predominantly target the phonemic and lexical levels: pronunciation accuracy, oral reading fluency, and decoding. The unit of assessment is the word or phoneme, and the reference is a known target text. The task addressed here has no reference text — the child is generating novel discourse — so alignment-based scoring does not apply.

2.3 Clinical speech therapy

Clinical instruments assess articulation, fluency disorders and language development against normative data, administered or supervised by a clinician. The purpose is diagnostic. The present system makes no diagnostic claim and is not a clinical instrument.

2.4 Why adult systems cannot be reskinned

Three specific obstacles, each corresponding to a constraint in Section 1.2.

First, **acoustic thresholds do not transfer**. A criterion calibrated on adult fundamental frequency will misclassify a seven-year-old systematically rather than randomly.

Second, **the scored construct is different**. Adult tools optimise delivery for speakers who already reliably generate content. In the target population the dominant structural failure is content-side — most visibly, exploring one idea versus enumerating several and abandoning them. Section 6 describes the rubric that follows from this.

Third, **the reporting contract is different**. An adult presented with 43/100 may treat it as actionable. The design premise in Section 1.2 holds that a child may instead stop. This is not a skin-level difference; it changes what the system is permitted to output, and therefore what it is worth computing.

3. Design principles

3.1 Delivery is measured, never inferred. A language model receives text. It cannot hear the audio. Asking it how a speaker sounded produces fluent text uncorrelated with the signal. All delivery quantities are therefore computed in code from timestamps and samples.

3.2 Determinism where a longitudinal claim is made. The parent-facing interface plots metrics across sessions. A trend line is only interpretable if re-analysing the same recording yields the same value. Every quantity plotted longitudinally is deterministic by construction.

3.3 Within-subject acoustic normalisation. Pitch and amplitude are interpreted only relative to the same speaker's own prior sessions, and are suppressed until that baseline exists.

3.4 Bounded corrective load. Exactly one corrective action is returned per session, selected by a fixed ordering in code.

3.5 Graceful degradation. Content assessment is the only component that can fail opaquely. Its failure path is explicit: the session degrades to delivery-only feedback rather than emitting fabricated scores.

4. System architecture

The implementation is a single Python service (FastAPI) with server-rendered templates. The modules below are as they exist in the repository; line counts are given to indicate where complexity sits.

```
audio (16 kHz mono PCM, browser-encoded)
|
v
```



The ordering in `pipeline.py` is explicit and commented: transcript, then deterministic delivery metrics, then baseline comparison, then content scoring, then feedback synthesis, then persistence.

5. Deterministic delivery measurement

5.1 Transcription

Transcription uses Groq's hosted `whisper-large-v3` (configurable via `WHISPER_MODEL`). The request sets `response_format="verbose_json"`, `timestamp_granularities=["word", "segment"]`, `language="en"` and `temperature=0.0`. Word-level timestamps are requested and are load-bearing: the pause and rate metrics are derived from them. The client filters malformed word entries rather than discarding the response, and returns an explicit words list so that a caller can detect their absence rather than silently scoring an empty sequence.

5.2 Temporal metrics

Speaking rate. Rate is computed over the *speaking span* — from the start timestamp of the first word to the end timestamp of the last — not over the clip duration. Where fewer than two words are returned, the clip duration is used as a fallback. The token count is obtained by regular-expression tokenisation of the transcript (`[a-z ' ]+`), falling back to the word-timestamp count.

Bands are assigned by these thresholds, as written:

Condition	Band	Flagged
<code>wpm < 90</code>	hesitant	yes

Condition	Band	Flagged
$90 \leq \text{wpm} < 100$	slightly slow	no
$100 \leq \text{wpm} \leq 150$	typical	no
$150 < \text{wpm} \leq 170$	slightly fast	no
$\text{wpm} > 170$	rushing	yes

Only hesitant and rushing set the flag consumed by corrective selection.

Hesitation pauses. For each adjacent word pair the inter-word gap is $\text{start}[n+1] - \text{end}[n]$. The gap is rounded to three decimal places *before* comparison, because word timestamps arrive at 10 ms resolution and raw floating-point subtraction turns a clean 0.70 s gap into 0.7000000000000002, which would classify identical pauses inconsistently across runs. A gap strictly greater than **0.7 s** is recorded as a hesitation.

Pause *distribution* is computed, not only pause count. The module returns: total count; longest gap; count falling within the first **15 s** of the speaking span; a three-way split of gaps into first, middle and final thirds of the span; and the first forty individual gap positions and durations. Two derived predicates follow: `weak_opening` when three or more gaps fall in the opening window, and `went_blank` when the longest gap reaches **3.0 s**.

Completion. Coverage is $\text{spoken_until} / \text{target_seconds}$. `ran_out` is asserted when coverage is below 0.6 or trailing silence after the final word exceeds 5 s. `went_full_distance` requires coverage ≥ 0.9 .

5.3 Disfluency

The filler lexicon is a nine-item tuple, exactly:

```
("um", "uh", "er", "like", "so", "basically", "actually", "matlab", "yaani")
```

The final two are Hindi discourse markers, included because the intended deployment population code-switches. Matching is whole-token and case-insensitive, performed after the same tokenisation used for rate. The module returns raw count, per-minute rate over the speaking span, a per-token breakdown, the modal filler, and a heavy predicate at ≥ 6 per minute.

Critically, the module also returns `reliable`, defined as `total > 0`. Whisper decoding may normalise disfluencies away. A zero count is therefore treated as weak evidence of absence rather than proof, and this propagates: the structured block passed to the phrasing model states "*Fillers: none detected (the transcriber may have dropped them — do not praise this)*" rather than reporting zero. Corrective selection additionally requires `reliable` before it will flag fillers. This is a mitigation, not a solution; see Section 10.3.

5.4 Repetition and lexical range

Both are implemented.

Repetition counts bigram and trigram types occurring twice or more. N-grams composed entirely of a 32-item stopword set are excluded, so that function-word sequences such as "and the" do not dominate. A rate is computed as $\Sigma(\text{count} - 1) \times 2 / \text{total_tokens}$. The heavy predicate fires at three or more repeated trigram types, or a rate above 0.25.

Lexical range reports type-token ratio over the whole transcript, together with a count of unique tokens absent from the shipped 1,046-word common list and longer than three characters. Note that type-token ratio is length-dependent; no length correction is applied (Section 10.6).

5.5 Acoustic measurement

Acoustic analysis exists and runs locally on the server; no audio is sent to any third party for this stage. WAV decoding uses the standard library wave module into NumPy arrays.

Amplitude. Root-mean-square energy is computed over 50 ms frames. Frames below 15% of peak RMS are treated as silence and excluded. The module returns mean RMS over speech frames, a scale-free coefficient of variation (`std / mean`), and the percentage drop between the first two-thirds and the final third of speech frames. `trails_off` is asserted at a drop of 25% or more. A separate speech-to-silence ratio thresholds frames at half the median RMS.

Fundamental frequency. F0 is computed with **Praat** via the `praat-parselmouth` binding, using `Sound.to_pitch()` with `pitch_floor=75.0` and `pitch_ceiling=600.0`. Unvoiced frames ($F0 = 0$) are discarded. If fewer than ten voiced frames survive, the module reports unavailability rather than a value. It returns mean, standard deviation, minimum and maximum F0, and the voiced frame count. Praat exceptions on very short or silent input are caught and reported as unavailable.

The 600 Hz ceiling is a documented inconsistency in the implementation: the source comment asserts that young children routinely exceed it, yet the ceiling is set at that value, which would clip such speakers. This is recorded as a divergence and as a limitation (Section 10.7).

5.6 Within-subject normalisation

The baseline module implements Principle 3.3. A speaker's baseline is the arithmetic mean of the **first three** sessions (`BASELINE_SESSIONS = 3`) for which the relevant quantity is available, taken oldest-first. Two quantities are baselined: F0 **standard deviation** (as the proxy for expressive variation) and **mean RMS**. Mean F0 is stored in the baseline structure but is not used for comparison.

Current-session values are expressed as ratios to that baseline and mapped to categories:

Quantity	Ratio condition	Status
F0 std	≤ 0.7	monotone
F0 std	≥ 1.3	expressive
F0 std	otherwise	typical
Mean RMS	≤ 0.7	quieter
Mean RMS	≥ 1.4	louder
Mean RMS	otherwise	typical

Before three qualifying sessions exist, `compare()` returns `calibrating = True` with both statuses set to `calibrating` and both ratios `None`, and **returns before computing any verdict**. The consequences are enforced at two further points. The text passed to the phrasing model becomes a single line — *"Pitch and volume: still calibrating (n/3 sessions) — do not comment on these"* — and the corrective selector guards both acoustic candidates behind not `calibrating`. An acoustic finding therefore cannot reach the child before the baseline exists.

6. Content assessment

6.1 Rubric

The model receives the transcript, the topic, the mode and the age band. It is instructed that the transcript is ASR output and to judge the thinking rather than the transcription, and it is instructed explicitly: *"You must NOT comment on voice, tone, pace, volume, confidence or nerves. You cannot hear the audio. Those are measured separately."*

Seven dimensions are scored on an integer scale of **1 to 5**, anchored in the prompt as 1 = absent, 3 = present but ordinary, 5 = strong for a child of this age. The dimension names and definitions are as written in the prompt:

Dimension	Definition as given to the model
specificity	concrete details, named people, places, numbers, moments, versus abstractions
personal_stake	did the child put themselves in it, an experience or an opinion they own
reasoning	claims backed with a reason ("because..."), versus asserted and dropped
angle	their own way into the topic, versus the most obvious take
development	one idea properly explored, versus several ideas touched and abandoned
on_topic	did they stay with the topic they were given
structure	a clear opening and a real ending, versus starting mid-thought and just stopping

The model additionally returns four structured observations — `has_opening`, `has_ending`, `distinct_points`, `used_example` — and three short free-text fields: `main_idea`, `strongest_moment`, and `biggest_opening`. The last is used by the feedback layer as raw material for an additive correction.

The rubric is **authored, not derived** from a corpus of judged speeches. Its construct validity is unestablished (Section 10.2).

6.2 Mode-dependent weighting

The two modes are scored differently at three distinct points in the code.

Weighted aggregate. A weighted mean over the seven dimensions produces an overall score, normalised to 0–100 as $(\text{mean} - 1) / 4 \times 100$. Weights differ by mode:

Dimension	Prepared	Impromptu
specificity	1.0	1.0
personal_stake	0.8	1.0
reasoning	1.0	0.8
angle	0.8	0.8
development	1.0	0.5
on_topic	1.0	1.0
structure	1.0	0.4

Development and structure are discounted in impromptu mode. The rationale encoded here is that in prepared mode these represent preparation failures, while in impromptu mode they are the ordinary cost of composing under time pressure.

Corrective threshold. The threshold at which thin development becomes eligible as a corrective action is mode-dependent: `development` ≤ 3 in prepared mode, ≤ 2 in impromptu mode.

Structure predicate. In prepared mode, a missing ending alone makes structure eligible. In impromptu mode, both a missing ending *and* a missing opening are required.

This aggregate is never shown to the child. It appears only in the parent-facing interface and in the stored record.

6.3 Robustness

The model is called with `temperature=0.0` and `response_format={"type": "json_object"}`. Because the configured default model is a reasoning model (openai/gpt-oss-120b), `reasoning_effort` is set to `low`; without it the model consumes the token budget on reasoning and returns no document.

Parsing is defensive and layered. `_extract_json` first strips markdown fences, attempts a direct parse, and on failure extracts the substring between the first `{` and the last `}` and retries. `_normalise` clamps every dimension into `[1, 5]`, rejects the response outright if **any** dimension is missing or non-numeric, bounds `distinct_points` to `[0, 10]`, and truncates the free-text fields.

On unparseable output the call is retried **exactly once**. If the second attempt also fails to parse, the function returns `{"available": False}` and the session proceeds with delivery-only feedback. Note an asymmetry: a *transport or API* exception breaks the retry loop immediately rather than retrying, so the retry budget applies to malformed documents, not to network failures. No default or imputed scores are ever substituted.

7. Feedback synthesis

7.1 Division of labour

The selection of *what to say* is performed in code. The model performs only *phrasing*.

`choose_focus()` evaluates nine boolean candidate conditions and returns the first that holds under this fixed ordering:

```

ran_out → rushing → fillers → development → specifics
        → monotone → trailing_volume → structure → repetition

```

with a `keep_going` fallback when none holds. The ordering is a literal tuple in the source and is evaluated by a `for` loop; **the model has no influence over which corrective action is selected**. Ordering rationale is authored, reflecting a judgement that a speaker who runs out of content cannot benefit from advice about vocal variety. This ordering has not been empirically validated (Section 10.4).

`choose_win()` selects the strength to praise from a ranked list of evidence-backed candidates, preferring a quotation of something the child actually said (`strongest_moment`) over generic categories. It then removes any candidate that would contradict the selected correction — praising steady pace while correcting rushing, praising low fillers while correcting fillers, and so on — via an explicit conflict map.

`metric_block()` assembles the structured facts: age band, mode, topic, transcript, duration against target, rate with band, filler summary or the unreliability notice, pause count with opening-window count and longest gap, the baseline-permitted acoustic lines, and the seven dimension scores with the three free-text observations. The phrasing model receives this block *plus the already-chosen* focus and win, and is instructed to write about those only.

If the phrasing call fails or returns unusable output, a deterministic template-based fallback produces both sentences from the same selected focus and win, and the stored record marks `source: "fallback"`. Feedback generation therefore has no single point of failure that can block a session.

7.2 Single-action constraint

The phrasing prompt states: *"Exactly ONE thing to try next time. Never two. Never a list."* The constraint is additionally enforced structurally — the function returns a two-field object (`win`, `tip`), and the selection layer supplies exactly one focus key.

The prompt further requires that corrections be phrased additively — what to add next time — rather than as a verdict on what was deficient. Age-band-specific style constraints are supplied, including sentence-length limits and, for the youngest band, an explicit list of abstract vocabulary to avoid.

7.3 Score withholding

The phrasing prompt's first absolute rule is: *"NEVER give a score, mark, rating, percentage, grade or any 'out of number'."* Counts of real events and durations are permitted; evaluative numbers are not.

This is an assessment design decision rather than a presentation preference. The system's objective function is practice volume, because improvement is conditioned on repeated practice. A design that measures accurately while suppressing the behaviour it exists to encourage has optimised the wrong quantity. The full numeric record — every dimension score, the weighted aggregate, all delivery metrics and the transcript — is computed, stored, and exposed to the account holder. Nothing is discarded; the numbers are routed away from the speaker.

We state plainly that the premise underlying this decision — that numeric feedback suppresses practice frequency in this population — is asserted from reasoning and is not measured within this work. It is the single most important claim in the paper that an evaluation should test (Section 9.4).

8. Data protection

Each control below is stated as implemented or planned, according to what is present in the source.

8.1 Implemented

Consent capture. Consent is stored per child, not per account, in two columns on the children table: `consent_at` (ISO 8601 UTC timestamp) and `consent_name` (the consenting adult's name, truncated to 120 characters).

Consent enforcement. The single request handler capable of initiating a recording checks `child["consent_at"]` and returns HTTP 403 when it is absent, before the uploaded bytes are read or forwarded anywhere.

Audio deletion timing. The uploaded audio is written to a scratch file because the Praat binding reads from disk. `pipeline.process()` wraps the entire analysis in `try / finally` and calls `unlink(missing_ok=True)` in the `finally` clause, so deletion occurs on every path including every error path. The session row is written with

audio_path = None. No recording outlives the request that produced it.

What is retained server-side. Per session: the full transcript text; the complete word-level timestamp array; all computed delivery metrics; the content scores; the generated feedback; mode, age band, topic, target duration and measured duration. Per child: a first name, an age band, and the consent record. **The transcript is retained in full and is a verbatim record of the child's speech.** This is a substantive retention, and is stated here rather than minimised.

Playback copies. The playback copy of the recording is held client-side in IndexedDB in the family's own browser. It is per-device and is not synchronised.

Deletion paths. Two are implemented: `delete_child()` removes one child and all dependent rows within an explicit transaction; `delete_all_for_parent()` iterates children and then removes the account.

Third-party scripts. Verified by inspection of every served template and static asset: there are no third-party script tags, no analytics, no advertising SDK, and no external font or CDN reference. The only external URLs appearing in markup are the authors' own profile links in the footer, plus the W3C SVG namespace URI, which is an XML identifier and not a network fetch.

8.2 Third parties in the data path

Two, both server-side and both named on the consent screen.

Groq receives the audio bytes for transcription and the transcript for content scoring. **Google** is the identity provider for the parent account and receives no data about the child.

8.3 Not implemented

Inference-provider retention terms are not configured in code. No zero-data-retention header, flag or account-level retention setting is set by the application when calling the inference provider. We searched the source for any such configuration and found none. Whatever retention policy applies is the provider's account default, not a property this system asserts. Configuring and verifying this is **planned, not implemented**, and no claim about the provider's handling of transmitted audio or transcripts is made in this paper.

Data protection legislation. The consent mechanism was designed with India's DPDP Act 2023 in mind, and the code comment records that intent. This paper makes no claim of legal compliance; no assessment against the Act's requirements for verifiable parental consent has been carried out.

9. Evaluation plan

No evaluation has been performed. Nothing in this section is a result. The system is implemented and operational; its accuracy, agreement with human judgement, and behavioural effects are all unmeasured. This section states what we intend to do, in the future tense.

9.1 Transcription accuracy on the target population

We will collect recordings from children within the three age bands, speaking freely for the durations the system supports, and will transcribe each by hand to produce a reference. We will compute word error rate against the system's transcripts, reporting per-band figures separately, since we expect degradation to be strongest in the youngest band. We will additionally measure whether disfluency tokens survive decoding, because the filler metric depends on it and currently protects itself only by declining to assert absence (Section 5.3).

Because every content judgement is conditioned on transcript quality, this measurement gates the interpretation of all others. Should error rates prove high, the appropriate response may be to narrow the supported age range rather than to adjust downstream components.

9.2 Agreement with human raters

We will recruit experienced elocution and debate judges, have them score a common set of recordings against the seven-dimension rubric, and compute inter-rater reliability among them before comparing the system to their consensus. Reporting system-human agreement without first establishing human-human agreement would make the comparison uninterpretable. We will report per-dimension agreement, as we expect on_topic to be considerably more tractable than angle.

9.3 Test–retest determinism

We will re-analyse identical recordings across repeated runs and confirm that every quantity plotted longitudinally is bit-identical, and that content scores vary only within the bounds permitted by decoding at temperature zero. This verifies Principle 3.2 as an implementation property.

9.4 Behavioural effect of score withholding

This tests the premise of Section 7.3 and is the study we consider most important. We will compare practice frequency and session-completion rates between the current design and a variant that exposes the numeric aggregate to the child. The outcome measure is return rate over a fixed window, not self-reported preference. We note that this design requires care with informed consent, since one arm deliberately exposes children to a condition we hypothesise to be discouraging.

9.5 Baseline sufficiency

We will test whether three sessions is adequate to characterise a speaker's acoustic norm, by computing the stability of the baseline estimate as sessions accumulate and determining the point at which additional sessions cease to move it materially. `BASELINE_SESSIONS = 3` is currently an assumption.

9.6 Corrective ordering

We will test whether the authored priority ordering in Section 7.1 corresponds to what coaches would prioritise for the same recordings, by presenting judges with sessions and asking for their single highest-priority correction.

10. Limitations

10.1 No empirical validation. Nothing about this system's accuracy or effect has been measured. Every performance claim that could be made is at present unsupported. This is the governing limitation and conditions all others.

10.2 The rubric is authored, not derived. The seven dimensions were written from reasoning about what distinguishes stronger from weaker speeches in this population, not induced from a corpus of speeches with known judge scores. The dimensions may not be mutually independent; `specificity` and `personal_stake` plausibly correlate. The weightings are likewise authored. No factor analysis or construct validation has been performed.

10.3 Filler detection depends on ASR decoding behaviour. Whisper may normalise disfluencies out of its output. The system mitigates this by treating a zero count as unreliable, propagating that caveat to the phrasing model, and refusing to select fillers as a corrective action when the count is zero. The mitigation prevents a false claim of fluency; it does **not** recover the information. Under-reporting is therefore possible and its magnitude is unknown.

10.4 The corrective ordering is asserted. The ordering encodes a plausible pedagogical claim but is not derived from measurement.

10.5 English only; code-switching unhandled. Transcription is requested with `language="en"`. The filler lexicon includes two Hindi discourse markers, which acknowledges that the population code-switches, but the system has no general mechanism for handling mixed-language speech: content scoring, tokenisation, lexical rarity and the common-word list are English-only. A child switching languages mid-speech will be transcribed and assessed unpredictably.

10.6 Lexical measures are length-sensitive. Type-token ratio decreases with transcript length by construction, so it is not comparable across sessions of different durations. No moving-average or length-corrected variant is implemented. The rarity count is likewise a raw count against a fixed 1,046-word list, which is small and was assembled by the authors rather than drawn from a frequency corpus.

10.7 Acoustic parameters are partly inconsistent. The Praat pitch ceiling is set to 600 Hz while the accompanying source comment asserts that young children routinely exceed it. Speakers above the ceiling will have F0 estimates clipped or dropped, which affects precisely the youngest band the normalisation scheme exists to protect.

10.8 Single speaker only. The system assumes exactly one speaker throughout the recording. No diarisation is performed and none is present in the source. Background speech, an adult prompting the child, or two children speaking will be attributed to the single speaker without detection.

10.9 Content assessment inherits transcription error. The prompt instructs the model to judge thinking rather than transcription, which is a mitigation of presentation but not of information loss. Content lost or corrupted in transcription cannot be assessed.

10.10 Playback copies are per-device. Because the playback recording is held in browser storage, it is unavailable on other devices and is lost if site data is cleared. This is a deliberate trade against server-side retention, and it is a real functional limitation.

10.11 No clinical claim. The system is not a diagnostic instrument and its output must not be read as evidence about speech or language development.

11. Conclusion

We have described the architecture of a system that assesses children's short speeches and returns bounded coaching feedback. The design partitions assessment by what each mechanism can actually determine: quantities recoverable from the signal and its timestamps are computed deterministically in code, and only the assessment of ideas — for which no deterministic method exists — is delegated to a generative model, which is given the transcript alone and told explicitly that it cannot hear the audio.

Three consequences follow from that partition. Longitudinal reporting becomes defensible, because the plotted quantities are reproducible by construction. Acoustic measures become interpretable across a developmentally heterogeneous population, because they are expressed relative to each speaker's own prior sessions and withheld before that baseline exists. And the corrective action becomes auditable, because it is selected by a fixed ordering evaluated in code rather than by a model whose selection criteria cannot be inspected.

We have deliberately separated what the system computes from what it shows the child. The numeric record is complete and is available to the account holder; it is withheld from the speaker on the argument that a system whose purpose is to increase practice should not emit the signal most likely to end it.

The architecture is implemented and operational. It is not validated. We regard the transcription accuracy measurement of Section 9.1 and the behavioural study of Section 9.4 as the two results that would determine whether this design is sound, and we report neither.

References

The following are referenced as the established methods and instruments the implementation composes. This is a systems paper; the citations describe components, not evidence for claims about performance.

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